

1. Administrative & Study Identification

Full Study Title: A PHASE 1, OPEN-LABEL, TWO-PERIOD, CROSS-OVER STUDY TO EVALUATE DOSE PROPORTIONALITY OF ATIRMOCICLIB (PF-07220060) PHARMACOKINETICS WHEN ADMINISTERED UNDER FED CONDITIONS TO HEALTHY PARTICIPANTS **Short Title/Acronym:** Atirmociclib Dose Proportionality Study in Healthy Participants **Protocol Number:** PF-07220060-Phase1 **NCT Number:** NCT07215078 **Sponsor Name:** Pfizer Inc. **Investigational Product:** Atirmociclib (PF-07220060) **Phase of Development:** Phase 1 **Medical Monitor:** Pfizer Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective * **Primary Objective:** To evaluate the dose proportionality on the pharmacokinetics (PK) of atirmociclib when administered across various dose ranges (Doses A, B, C, and D) under fed conditions. * **Secondary Objectives:** To evaluate the safety and tolerability of single oral doses of atirmociclib in healthy volunteers.

2.2 Background & Rationale Atirmociclib is a highly selective CDK4 inhibitor in clinical development. The purpose of this Phase 1 clinical trial is to characterize how the drug moves through the body (pharmacokinetics) and to establish whether the drug's exposure increases proportionally as the dose increases when taken with a meal. Understanding dose proportionality under fed conditions is essential to define safe and effective dosing guidelines for future patient trials.

3. Study Design & Methodology

3.1 Design Framework * **Study Type:** Interventional * **Control Type:** Active Comparator (Cross-over design comparing different dose levels) * **Blinding:** Open-label * **Randomization:** Participants randomized to 1 of 12 treatment sequences among 6 cohorts

3.2 Planned Enrollment & Duration * **Target \$N\$:** 72 healthy participants * **Number of Sites:** Single/Local Clinical Research Unit * **Estimated Study Start:** Late 2025 / Early 2026 * **Estimated Primary Completion:** N/A (Short-duration Phase 1)

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Male and female participants aged 18 to 65 years. 2. Healthy as determined by medical evaluation, including medical history, physical examination, and laboratory tests. 3. BMI of 17.5-30.5 kg/m² and a total body weight >50 kg (110 lbs).

4.2 Exclusion Criteria 1. Any condition possibly affecting drug absorption (e.g., gastrectomy, cholecystectomy). 2. History of human immunodeficiency virus (HIV) infection, hepatitis B (HBsAg positive), or hepatitis C (HCVAb positive). 3. Concomitant use of any medications or substances that are strong inducers or inhibitors of CYP3A4 or UGT2B7 within 5 half-lives plus 14 days (up to 28 days) prior to the first dose of atirmociclib.

5. Intervention & Treatment Plan

5.1 Investigational Regimen * **Dosage/Frequency:** Single oral doses of Atirmociclib (Doses A, B, C, and D) assigned by sequence cohort. * **Route of Administration:** Oral (higher drug load IR MST tablets), given about 30 minutes after a moderate-fat standard calorie meal. * **Duration of Treatment:** Two confinement periods of 6 days each, with a minimum 7-day interval between periods. Total study duration up to 71 days including follow-up.

5.2 Concomitant & Prohibited Therapy * **Permitted:** Standard medical care, if non-interfering and approved by the investigator. * **Prohibited:** Strong CYP3A4 and UGT2B7 inhibitors or inducers.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
Screening	Day -28 to 0	Informed Consent, Medical History, Physical Exam, Safety Labs
Period 1	Day 1 to Day 6	Admission, Standard Calorie Meal, First Single Dose, Serial blood PK sampling up to 120 hours
Period 2	Minimum 7-day interval	Shuffle to alternate dose, Standard Meal, Single Dose, Serial PK sampling, Discharge on Day 6
End of Study	Day 28 to 35 post-dose	Safety follow-up assessment and final laboratory checks

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint * **Definition:** Dose proportionality on the pharmacokinetics (PK) of atirmociclib (including parameters like AUC and Cmax) evaluated across the dosed cohorts. * **Measurement Method:** Serial blood PK sampling taken up to 120 hours after administration.

7.2 Secondary & Exploratory Endpoints * Safety and tolerability profile across different oral dose levels of atirmociclib. * Incidence and severity of Treatment-Emergent Adverse Events (TEAEs).

8. Safety & Adverse Event Reporting

- **Adverse Event (AE) Monitoring:** Continuous clinical and physiological observation during the 6-day inpatient periods and final safety follow-up.
 - **Serious Adverse Events (SAEs):** Standard 24-hour reporting to the sponsor.
 - **Data Monitoring Committee (DMC):** Sponsor-led internal safety and PK review.
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9. Statistical Analysis Plan (SAP) Summary

- **Analysis Populations:** PK Concentration Set (participants with sufficient samples to determine AUC/Cmax) and Safety Analysis Set.
 - **Sample Size Justification:** $N = 72$ ensures adequate power for within-subject comparisons in a 2-period cross-over design across 6 sequences/cohorts.
 - **Handling of Missing Data:** Participants with insufficient PK data points are excluded from the primary PK calculations; safety variables evaluated based on observed data.
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10. Quality Assurance & Regulatory Compliance

- **GCP Compliance:** This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki.
 - **Monitoring Plan:** Continuous onsite monitoring during inpatient stay.
 - **Audit Trail:** Standard Phase 1 eCRF locking, source data verification (SDV), and data clarification forms.
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11. Study Identifiers & Governance

- **Primary ID:** PF-07220060-1
- **Registry Number:** NCT07215078
- **Sponsor/Lead Organization:** Pfizer Inc.
- **Study Title:** Study About Whether Atirmociclib/PF-07220060 Proportionally Increases Exposure as Dose Increases in Healthy Participants
- **Official Regulatory Title:** A PHASE 1, OPEN-LABEL, TWO-PERIOD, CROSS-OVER STUDY TO EVALUATE DOSE PROPORTIONALITY OF

12. Clinical Framework (Study Specific)

- **Primary Condition:** Healthy Volunteers
 - **Diagnosis Threshold:** Normal medical evaluation and BMI 17.5-30.5 kg/m²
 - **Medical Methodology:** Phase 1 Pharmacokinetics Evaluation
 - **Optimization Target:** Dose proportionality (AUC, Cmax) assessment
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13. Study Procedures & Methodology

- **Management Pathway:** Inpatient stay at a dedicated Phase 1 Clinical Research Unit
 - **Education Protocol (HCP):** Phase 1 protocol training and dosing administration
 - **Education Protocol (Patient):** Informed consent, study diet/fed-state requirements
 - **Quality Audit Mechanism:** Direct clinical observation, strict feeding schedules, and timed PK blood draws
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14. Observation & Follow-Up Schedule

- **Total Duration:** Up to 71 days maximum per participant
 - **Onsite Visit Cadence:** Two 6-day admission periods separated by a 7-day interval
 - **Remote Monitoring Frequency:** N/A (Inpatient setting)
 - **Checklist Review Interval:** Safety follow-up executed 28 to 35 days after the final dose
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15. Sign-off & Approval

Role	Name	Signature	Date	:--	:--	:--	:--
Principal Investigator	[Name]	____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Name]	____	[DD-MMM-YYYY]		Lead		
Statistician	[Name]	____	[DD-MMM-YYYY]				